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Coupled Level-Set/Volume-of-Fluid Method for Simulation of Injector Atomization

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This paper presents results of a multiphase computational fluid dynamics code using a coupled level-set/volume-of-fluid method to simulate liquid atomization. This interface-capturing approach combines the mass conservation properties of the volume-of-fluid method with the accurate surface reconstruction properties of the level-set method, and it includes surface tension as a volume force calculated with second-order accuracy. Developed by one of the authors, the multiphase code builds upon the combined level-set/volume-of-fluid methodology to enable bubbly flow, liquid breakup, and phase-change simulations. The extension presented in this paper couples a Lagrangian dispersed phase model for postbreakup tracking of droplets with block-structured adaptive mesh refinement on the Eulerian grid. Under an appropriate set of criteria, the transfer of droplets representation from the Eulerian to the Lagrangian discretization enables the simulation of sprays on larger domains and for longer physical times without degrading the overall accuracy of the calculation. A demonstration for like-on-like jet impingement of relevance to cold startup of liquid rocket engines is presented here at low, moderate, and high injection velocities, including validation with experimental data.

Nomenclature

d_d	=	Lagrangian particle diameter
d_0	=	injection orifice diameter
d_{10}	=	arithmetic mean of spray drop size
F	=	liquid volume fraction
f_a	=	$v_{\text{jet}}/\Delta l$ atomization frequency
f_s	=	sampling frequency
l_j	=	jet preimpingement length
m	=	Δx multiple defining the critical radius
p	=	pressure
R_{cri}	=	critical drop radius for Lagrangian transformation
Re	=	$\rho_L v_{\text{jet}}\, d_0/\mu_l$ Reynolds number based on liquid properties
Re_d	=	$\rho_G \lvert \mathbf{u}^c - \mathbf{v}_d \rvert d_d/\mu_G$ droplet Reynolds number
t	=	half-thickness of the impinging sheet
$\mathbf{u}$	=	Eulerian velocity field
V_{cri}	=	critical drop volume for Lagrangian transformation
V_L	=	blob volume
$\mathbf{v}_d$	=	Lagrangian particle velocity
v_{jet}	=	nominal injection velocity
We	=	$\rho_L v_{\text{jet}}^2 d_0/\sigma$ Weber number based on liquid properties
x_b	=	breakup length (intact liquid core length from the impingement point)
α	=	drop sphericity
β	=	number of droplets per unit volume
Δl	=	ligament spacing
Δx_f	=	finest grid spacing on the Cartesian mesh
Δx_c	=	coarsest or base grid spacing
μ	=	cell-averaged dynamic viscosity
ρ	=	cell-averaged density or gas-liquid density ratio in the linear analysis
σ	=	surface tension coefficient
ϕ	=	level-set function

I. Introduction

IMPINGING jet injectors rely on the mutual impact of two or more high-velocity jets to atomize the propellants. The elements may flow either a single propellant (like impinging) or fuel and oxidizer (unlike impinging), and typical geometries involve two, three, or five streams (from doublet, triplet, and pentad elements). This impingement results in a highly directional distribution of propellant(s) that must be managed to provide good mixing efficiency and combustor wall thermal compatibility. In the case of the doublet element, the distribution is that of a "fan" of preatomized liquid material, which is at right angles to the common plane of the two liquid streams. For like impinging doublets, these fans must be interwoven in the overall injection pattern to ensure sufficient mixing of fuel and oxidizer. Injection velocity and stream diameter control the droplet size distribution. The fans break up in various modes, but usually exhibit waves that separate into liquid ligaments in the plane of the fan [1]. These ligaments then fracture into droplets, typically through Rayleigh breakup.

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Because the atomization and mixing processes of such injectors are complex and strongly three-dimensional, impinging injector sprays have mainly been characterized experimentally, like most other similar atomization systems [2–4]. An example is the study carried out at the United Technologies Research Center High-Pressure Spray Facility [5], where the effects of liquid properties and chamber conditions on the spray characteristics were investigated using a Malvern droplet size analyzer. Reynolds numbers were between 7.3×10^4 and 8.4×10^4 for the oxidizer and between 1.7×10^4 and 1.9×10^5 for the fuel. Photographs of the injection in Ref. [5] displayed the two jets crossing each other's path at the impinging point and forming a dense spray in the common plane of the streams, suggesting a loss of cohesion of the streams before contact. Drop sizes from an Aerometrics phase Doppler particle analyzer (PDPA) were found to decrease with increasing liquid flow rate, decreasing surface tension and increasing chamber pressure.

The formation of a liquid sheet that ruptures in increasingly less coherent ligaments with higher jet velocity has been described in impinging water jet experiments by Anderson et al. [6] and Ryan et al. [7]. In their studies (from now on referred to with the acronym AR for ease of notation) the Weber number was increased from 350 to 6600, with Reynolds numbers between 2.8×10^3 and 2.6×10^4. The measured breakup length along the centerline showed a strong dependence on the turbulent condition of the jet at injection. In what AR refer to as turbulent impingement, the core breakup length and the ligament frequency correlated well with the stability parameter d_0/v_{jet}. These data, as well as the PDPA characterization of the spray fan, are used for validation of the simulations that are presented here.

Recently, interface-capturing simulation techniques have been applied to simulate the primary atomization of impinging jets. In a setup similar to the AR experiments [6,7], Inoue et al. [8,9] used a cubic-interpolated advection scheme coupled with hybrid levels set and a multi-interface advection and reconstruction solver. Simulations showed the formation of the characteristic fan-shaped sheet, which rapidly developed a flapping mode leading to its breakup. More recently, Li et al. [10] demonstrated the effectiveness of coupling the multiphase code based on the interface-capturing combined level-set/volume-of-fluid (CLSVOF) formulation [11] with the Lagrangian tracking of the spray. On a dynamically adaptive, block-structured grid, this approach selectively enables the highest grid density at the liquid interface, thereby increasing the accuracy of the primary breakup calculation. High-resolution grid boxes are embedded in a coarser (compared with the scales of primary atomization) computational domain for the Lagrangian representation of the dilute spray, thus enabling simulations like the one displayed in Fig. 1.

Highlights of the numerical method are provided in the next section. As the computational domain is sufficiently broad to include the dilute spray region, it becomes possible to carry out a validation study of jet impingement atomization with the PDPA measurements from the AR experiments [6,7]. This comparison, including the core length of the impinging sheet and the ligament spacing for a set of three injection velocities, constitutes the second part of the paper.

II. Numerical Method

The Navier–Stokes equations for incompressible flow of two immiscible fluids (such as a liquid/gas system) are solved with the one-fluid approach according to the level-set equations for multiphase flow [12]:

$$\rho(\phi)\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = -\nabla p + 2\nabla[\mu(\phi)\mathbf{D}] - \sigma\kappa\nabla H(\phi) \tag{1}$$

$$\nabla\mathbf{u} = 0 \tag{2}$$

$$H(\phi) = \begin{cases} 1 & \phi \geq 0 \\ 0 & \phi < 0 \end{cases} \tag{3}$$

$$\rho(\phi) = \rho_L H(\phi) + \rho_G[1 - H(\phi)] \tag{4}$$

Fig. 1 Simulation of impinging jets for the low-velocity injection case.

$$\mu(\phi) = \mu_L H(\phi) + \mu_G[1 - H(\phi)] \tag{5}$$

$$\frac{\mathrm{D}\phi}{\mathrm{D}t} = 0 \tag{6}$$

In the preceding equations, ϕ is the level-set function, $\mathbf{D}$ is the deformation tensor, $\mathbf{D} = [\nabla\mathbf{u} + (\nabla\mathbf{u})^T]/2$, H is the Heaviside function, and $\mathrm{D}/\mathrm{D}t$ is the material derivative. Buoyancy effects are neglected in this study and not included in the momentum equation.

In the CLSVOF method, the advection equation for the liquid volume fraction

$$\frac{\mathrm{D}F}{\mathrm{D}t} = 0 \tag{7}$$

is solved in step with ϕ so that the level-set can be reinitialized in a local mass preserving fix. In this fix, the volume fractions are used with the interface normals calculated from ϕ to construct a volume-preserving distance function. The smooth zero level of ϕ represents the time-evolving interface. Its curvature is calculated with second-order accuracy from the volume fractions by the method of height fractions. The properties of density ρ and dynamic viscosity μ are functions of ϕ everywhere in the computational domain.

The solution proceeds via the projection method for incompressible flow. The multigrid preconditioned conjugate gradient method introduced by Tatebe [13] is adopted for solving the sparse matrix system that results from discretizing the pressure correction equation with discontinuous coefficients. The velocity extrapolation from the liquid to the gas phase described in [11] enables large liquid-to-gas density ratios. As shown there, the method yields solutions in the zero gas density limit, which are comparable in accuracy with the second-order approach for a spatially constant pressure gas presented in [14]. Because the volume-of-fluid (VOF) function is advected using the liquid velocity field, the liquid velocity is numerically divergence free in the liquid region, but not necessarily divergence free in the gas region. This might cause small over- or undershoots in the VOF function. Also, the VOF advection operator is discretized in a conservative operator-split manner; it therefore allows for the VOF function to exceed one or fall below zero. However, "clipping" back to one or zero, respectively, was not allowed in the simulations presented here. Volume conservation tests indicate good conservation properties of the algorithm; one example will be presented later in this paper. A full account of the CLSVOF interface-capturing method and of the multiphase flow solver developed by Sussman

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et al. can be found elsewhere [11], together with verification studies on surface-tension-driven oscillations of a spherical drop and of ligament pinch-off induced by capillary instability.

With adaptive mesh refinement (AMR), the cells that are crossed by the liquid–gas interface can be tagged for grid refinement. Starting from the base level, boxes (with a maximum size of, say, 32^3 cells) are combined to cover all the tagged cells within assigned coverage efficiency. This set of blocks with the same grid spacing forms level 1. This level is in turn tagged for refinement at the interface, and the process is repeated until the input grid resolution is achieved. The refinement ratio between two consecutive refinement levels is two. During the simulation, the data on the fine level are either copied from a previous time step or, when the grid structure has changed locally, conservatively interpolated from the underlying coarse level. The interface, however, is always embedded in the finest grid level to avoid gross interpolation errors. In a time step, the calculation is carried out on all levels, and the updated data on a fine level are averaged to the underlying coarser one.

For a given minimum grid spacing, a higher granularity of the coverage (i.e., the prescription of many small AMR boxes) minimizes the grid count but maximizes data communication between boxes, and vice versa. Assuming the optimal coverage closely follows the interface, and is therefore essentially two-dimensional, it is only slightly optimistic to estimate that doubling the grid density in a three-dimensional simulation corresponds to increasing the storage and the execution time by a 2^3 factor. This estimate includes a 2^2 factor at the interface and a factor of 2 in time due to the convective time-step restriction. This 2^3 increase is compared with the 2^4 factor (three dimensions plus time) required by a subdivision of the entire computational grid. Ideally, this saving in resources is repeated at every new refinement level.

In the previously described computational framework, as the liquid jets fracture and the spray disperses, most of the computational domain tends to be overlayed by the highest refinement level. In the simulation, the saving afforded by AMR would eventually disappear and be overtaken by the computational overhead of administering the grid blocks. This is because, on an Eulerian grid, the transport of an individual drop requires a certain grid density even to conserve mass and advect the correct shape of the drop. This requirement would be increased by at least one order of magnitude by the inclusion, for instance, of surface evaporation. Direct numerical simulation of a fuel spray from injection to evaporation thus seems unfeasible in an Eulerian-only framework.

The solution proposed in this work is to couple the block-structured grids with a mechanism to convert sufficiently small, isolated lumps of liquids into Lagrangian particles. What is new in this approach is that, because such droplets are removed from the Eulerian description, the hierarchy of refinement levels reverts to the underlying base level in the region interested by the transformation. Thus, most of the computational resources can be localized in a compact volume near injection, where our current knowledge is more limited, while using a relatively coarse grid far from injection, where we can take advantage of engineering models of spray transport. Moreover, where the resolution becomes insufficient to capture the viscous effects away from the interface, existing subgrid scale models for single-phase flows could be used.

Code execution timing in a few representative cases indicates that remeshing costs less than 6% of the total CPU time after the simulation reaches steady state. The reason behind the low cost of block-structured AMR is that the algorithm is based on simple tagging, cell splitting, and data interpolation, unlike the conventional remeshing algorithms in which complicated calculation of new cell coordinates is required to ensure mesh quality. In the typical simulation run, we found that four tasks (pressure correction solve, level-set reinitialization, nonlinear advection, and velocity diffusion) took approximately 75% of CPU time. The remaining 25% of CPU time was absorbed by tasks like interlevel data communication and grid generation. This 75/25 split was found to be the same for the two- and three-level calculations (using between 64 and 256 processors), which will be discussed next.

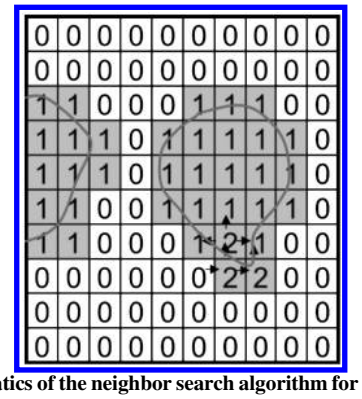

Fig. 2 Schematics of the neighbor search algorithm for liquid structure identification.

A. Droplet Identification

In this step, almost spherical droplets in dilute spray regions are converted into Lagrangian particles similar to the approach first proposed by Herrmann [15]. Because mesh refinement is based on the location of the liquid–gas interface, small droplets are always resolved by the finest refinement level, therefore, the identification procedure described next is performed exclusively on the finest level.

To find candidate droplets for the Lagrangian description, liquid structures separated by the gas phase are identified by the neighbor-searching algorithm illustrated in Fig. 2. The drawing is simplified to a two-dimensional representation, but the algorithm is fully three-dimensional. First, all the cells are tagged: $G = 1$ (color gray) in the liquid phase, including the liquid–gas interface; and $G = 0$ (color white) in the gas phase. Contiguous cells are then collected into separate cell lists to represent separate liquid structures. This is achieved by searching cells in the domain until a cell with $G = 1$ is found. Then, the cell tag is set to $G = 2$ and a new list is prepared for the liquid structure containing that first cell. Next, the immediate neighbors of the cell are searched. If a cell with $G = 1$ is found, its tag is set to $G = 2$ and it is added to the list of cells. The search is carried out in the way of a multibranch tree. The arrows in Fig. 2 indicate the search direction for the first few search steps. Eventually the searching is stopped when none of the neighbors of the cell within the list can be found with a tag $G = 1$, and thus we obtain one isolated liquid structure represented by a list of cell indices tagged by $G = 2$. The algorithm continues scanning the grid searching for other isolated liquid structures to be represented by separate lists of cells.

As the search is carried out, running sums of the volume, the center of mass, and the average velocity of the liquid structure are also updated. Note that the liquid–gas interfaces are contained in the obtained liquid structures (Fig. 2) and that the computation of the total liquid volume based on cell volume fraction is accurate. The updated liquid volume and center of mass are used to determine the eligibility of the liquid structure to be transformed into a Lagrangian particle, as described next. For instance, the liquid structure can be disqualified early if the running sum of volume is greater than a user-defined critical volume. No further operation, except for searching neighbors and tagging, is performed for the remaining cells. Such a procedure greatly reduces the algorithm cost when searching for liquid structures with large volumes.

The parallelization of the identification procedure when neighboring boxes are on different processors is very simple because there is no communication of liquid structure information between blocks on different processors. In Herrmann's approach [15], a complex multiblock identification procedure is adopted to resolve the issue of liquid structures being separated by different blocks because of domain decomposition. Liquid structures are identified independently in each block and then merged together by assigning a unique identifier. We avoid this complexity by making use of the dynamic features of block-structured AMR. Particularly, any liquid structure that is cut by the boundary of boxes (blocks) is disqualified, and only the structures that are completely contained by a box are considered for transformation. This procedure might seem arbitrary, but, in

reality, mesh refinement needs to be performed adaptively at every time step, so that a disqualified liquid structure has a very good chance to be completely enclosed by a new box within a few iterations.

B. Conversion Criteria from Eulerian to Lagrangian Representation

The removal of liquid structures from the Eulerian description is based on criteria of droplet size, sphericity, and local droplet concentration. The volume of a candidate liquid structure is constrained to be less than or equal to a critical value [15],

$$V_L \leq V_{\text{cri}} = \frac{4}{3}\pi R_{\text{cri}}^3 \tag{8}$$

where $R_{\text{cri}} = m\Delta x$ and m is a user-prescribed small integer. If the droplet is small enough, further breakup (secondary breakup) cannot occur due to the high surface tension compared with the aerodynamic force. In the calculations, the Weber number of each particle is monitored at each time step, and this diagnostic confirms the absence of the conditions for secondary breakup in the simulations presented here. The maximum value of m is constrained by the requirement that the Lagrangian particle must be contained in a cell of the base grid.

A second criterion is the droplet sphericity, defined as the nondimensional quantity [15]

$$\alpha = \frac{R_{\max}}{\max[\Delta x, (\frac{3}{4\pi}V_L)^{1/3}]} \tag{9}$$

$R_{\max}$ is the maximum distance from the center of mass of the liquid blob to the centroid of any of its cells; V_L is the blob volume. Because largely deformed blobs are more probable to experience breakup, which we are equipped to solve in the Eulerian representation, the transformation to a spherical particle can only occur only when $\alpha \leq \alpha_{\text{cri}}$.

An example of the effect of size and sphericity criteria is illustrated in Fig. 3 for a single jet subject to a gas crossflow in the direction perpendicular to the liquid injection direction; the reader is referred to [10] for additional details. The objective is to illustrate how the small, quasi-spherical droplets can be identified. In this example, we apply either size, sphericity, or both criteria to identify the liquid structure to be transformed and show the remaining liquid phase after transformation under the different criteria. In Fig. 3c, only the size restriction is enforced: $R_{\text{cri}} = 5\Delta x$. Compared with Fig. 3a, in which no transformation is performed, a large number of small ligaments are removed from the Eulerian description irrespective of their shapes. In Fig. 3d, only the sphericity condition is enforced: $\alpha_{\text{cri}} = 3$. Much of the highly deformed ligaments remain intact because of their stretched shape. The jet column, however, meets the sphericity criteria and therefore is removed from the Eulerian description. The correct situation is when both size and sphericity restrictions are enforced, as in Fig. 3b: $R_{\text{cri}} = 5\Delta x$ and $\alpha_{\text{cri}} = 3$. Only small and nearly spherical droplets away from the jet column are transformed. This result indicates the importance of applying both criteria of sphericity and size.

A maximum local spray concentration β_{cri} is also defined, below which the transformation to the Lagrangian description can occur. Once generated from its Eulerian representation, each droplet is

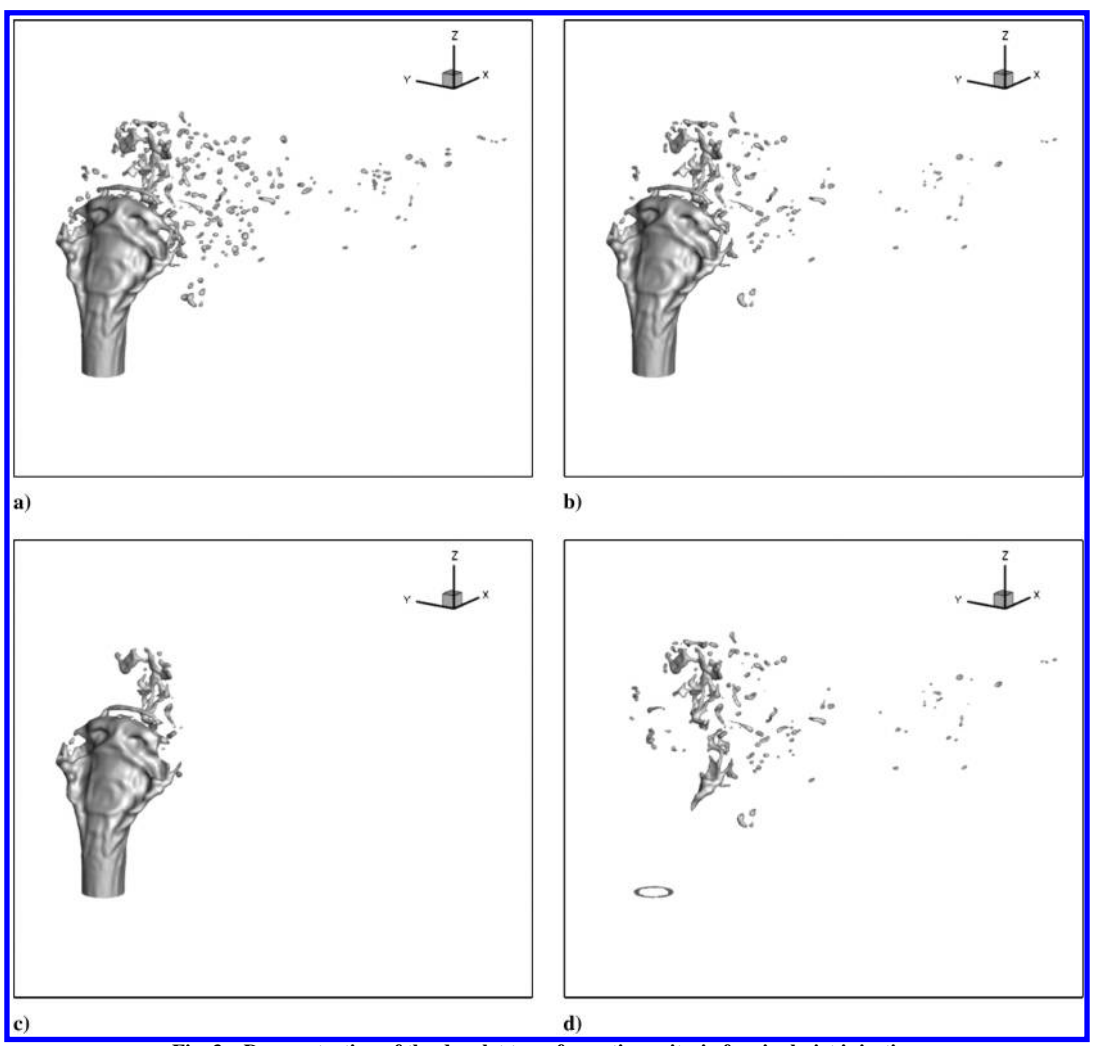

Fig. 3 Demonstration of the droplet transformation criteria for single-jet injection.

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"frozen" into a spherical shape, so that breakup, merging, or even droplet deformation are not allowed. This convenient modeling choice is not of universal applicability, but it is consistent with a situation in which the spray is expected to become increasingly diluted. The parameter β_{cri} should therefore be chosen to ensure that droplet collision is unlikely. The local concentration is defined as the ratio of the total liquid volume in an AMR box to the volume of the box. For the cases examined in Sec. III, we observed limited sensitivity of the results to variations of this parameter if $\beta_{\mathrm{cri}} \sim 0.01$.

C. Lagrangian Droplet Generation and Transport

At the end of the selection process, the liquid structures that meet all the criteria described in the previous section are removed from the Eulerian representation by setting the level set ϕ and volume fraction F of the cells there as

$$\phi = -|\phi| \quad \text{and} \quad F = 0 \tag{10}$$

While this operation is performed, the droplet volume, center of mass, and velocity are calculated as

$$V_d = \int_d \mathrm{d}x \qquad \mathbf{x}_d = \frac{1}{V_d}\int_d \mathbf{x}\,\mathrm{d}x \qquad \mathbf{v}_d = \frac{1}{V_d}\int_d \mathbf{u}^f\,\mathrm{d}x \tag{11}$$

and stored in a particle list. Thus, a drop spanning a few grid cells on the finest grid level is reduced to a pointlike Lagrangian particle with position $\mathbf{x}_d$ and velocity $\mathbf{v}_d$ on the base, or coarsest, level.

The liquid mass is rigorously preserved in the execution of the conversion. An example of this is displayed in Fig. 4 for one of the simulations presented later ($v_{\mathrm{jet}} = 18.5$ m/s) with a fine grid size of $\Delta x_f = 31.25$ μm. Because the flow rate through the two orifices is constant, the total liquid mass that gets introduced in the computational domain should be linear with time until droplets begin to exit the domain. In the time interval displayed in the plot, the amount of liquid volume represented by CLSVOF on the AMR grid has reached a stationary value where the inflow mass is on average balanced by the mass of newly created Lagrangian drops. The overall volume of the liquid phase in the Eulerian representation as a function of time corresponds to the sequence of triangles in Fig. 4. In the same diagram, the volume of all the droplets in the Lagrangian representation that are contained in the computational domain is shown by the sequence of square symbols, and the time history of the volume of all the Lagrangian droplets that have left the domain is shown by the diamond symbols. If these three components are summed at each time step, the result is an approximately straight line (circle symbols), with deviations from linear growth that are less than 1% of the overall liquid volume at time 2.5 ms. This result proves the excellent convergence properties of CLSVOF, as well as the correct implementation of the transformation algorithm in the liquid phase. In the gas phase, the algorithm forces the volume fraction of the gas phase to one in cells where new Lagrangian droplets are created. The added gas mass is a small fraction of the spray mass because of the large liquid–gas density ratio considered, but the value of a more rigorous implementation of the transformation algorithm is noted.

The particle trajectory is calculated by solving the differential equations

$$\frac{\mathrm{d}\mathbf{v}_d}{\mathrm{d}t} = F_d(\mathbf{u}^c - \mathbf{v}_d) \quad \text{and} \quad \frac{\mathrm{d}\mathbf{x}_d}{\mathrm{d}t} = \mathbf{v}_d \tag{12}$$

for all the elements of the droplet list. Note that the gas flow velocity $\mathbf{u}^c$ is obtained by linear interpolation at the instantaneous droplet location from neighboring cell values at the coarsest level, whereas the initial value of $\mathbf{v}_d$ is evaluated from the Eulerian flowfield $\mathbf{u}^f$ on the finest level of refinement. The drag F_d is calculated with a simple model [16] for a rigid sphere of diameter d_d,

$$F_d = \frac{18\mu_G}{\rho_L d_d^2}\frac{C_D Re_d}{24} \tag{13}$$

The drag coefficient is $C_D = 0.424$ for $Re_d = \rho_g|\mathrm{u}^c - \mathrm{v}_d|d_d/\mu_G > 1000$ and

$$C_D = \frac{24}{Re_d}\left(1 + \frac{1}{6}Re_d^{2/3}\right) \tag{14}$$

otherwise.

It should be noted that the introduction of the drag relation introduces an empirical model, albeit well accepted by the spray community, in a calculation that is otherwise devoid of tunable coefficients. The gas phase is furthermore assumed not to be impacted by the motion of the Lagrangian particles; that is, only one-way coupling is used in these simulations. This simplification can be justified in the case of impinging jets in a still gas when most of the momentum transfer (from the jets to the gas phase) occurs in the dense spray, where the liquid phase is Eulerian. The assumption is convenient in the current study because droplet size validation is carried out by collecting spray samples over a relatively wide region of space. Two-way Eulerian–Lagrangian coupling has been implemented and tested in other applications, such as atomization by a crossflowing gas [17].

To verify the consistence of the transformation algorithm and of the Lagrangian transport, a snapshot from a simulation with Lagrangian particles (rendered as spheres with the appropriate diameter) is compared in Fig. 5 with a fully Eulerian calculation. The simulation continues the test case of Fig. 3 and consists of a single injection with a gas crossflow that accelerates the droplets in the transverse direction. Before the switch, the simulation was carried out for 1000 time steps. The transformation was introduced after that point only for Fig. 5b, then both cases were continued for 100 steps. The positions of the Lagrangian particles after 100 time steps match the positions of the drops in Fig. 5a, confirming the accuracy of the transformation and the adequacy of the Lagrangian transport model. Naturally, the grid count in Fig. 5b is significantly reduced, as one can see from the distribution of grid blocks, compared with the case in

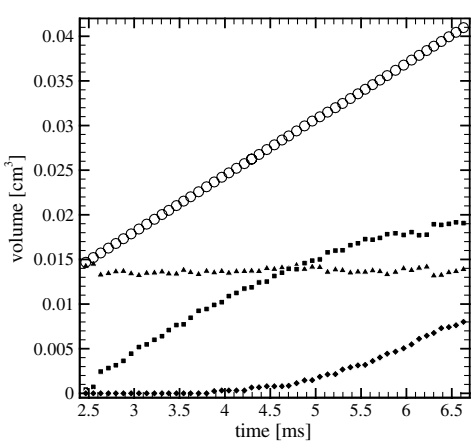

Fig. 4 Liquid volume variation as a function of time.

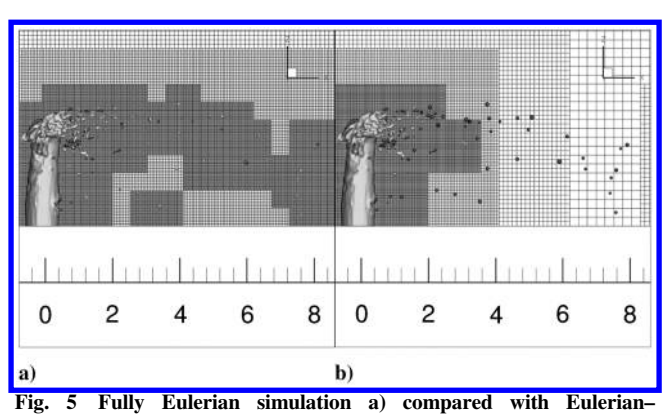

Fig. 5 Fully Eulerian simulation a) compared with Eulerian–Lagrangian transformation after 100 time steps b).

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which all droplets are resolved using all the available refinement levels.

III. Results

The AR experiments used precision bore glass tubes to minimize the effects of surface roughness [6,7]. The internal diameter of the tubes was $d_0 = 0.635$ mm, the impingement angle 60 deg, and the preimpingement length $l_j = 25.4$ mm. PDPA measurements of diameter distribution at the spray fan centerline were reported at 41 mm downstream of the impingement point. The PDPA optical configuration was set to provide a 40–1400 μm range of drop diameters. The test liquid was water $\rho_l = 998\ \mathrm{kg/m^3}$, surface tension $\sigma = 0.076$ N/m, and dynamic viscosity $\mu_l = 0.0010\ \mathrm{kg/(m \cdot s)}$ in air at ambient conditions. These same properties are used in the simulation: The liquid–gas density ratio is 832 and the dynamic viscosity ratio is 50. In addition to the PDPA measurements, images of the impinging sheet were taken with diffuse backlighting provided by a strobe light with approximate duration of 5 μs. The length from the impingement point to the perceived breakup of the intact sheet and the distance between adjacent ligaments were estimated as averages extracted from several of these photographs.

A simulation snapshot is shown in Fig. 1 with the rendered zero level-set isosurface and with the Lagrangian particles in lighter color. These pointlike particles are given a spherical shape and rendered with an effective radius corresponding to the particle's volume. Injection occurs in the y-z plane, with the z axis aligned in the common direction of the two jets; see Fig. 6. To reduce the overall size of the computational domain, the bottom plane (at $z = 0$) is not the injection wall of the experiment, but rather truncates the two liquid columns so that their impingement occurs at $z = 4$ mm. Outflow boundary conditions are enforced on all the other planes. The overall box size is $12 \times 16 \times 24$ mm.

The injection velocity consists of a plug flow profile, without accounting for the relaxation due to shear at the interface when the liquid exits the pipe. Furthermore, the Reynolds number in the pipe for the cases under consideration ranges from 4000 to 12,000. Because the glass tubes have a length-to-diameter ratio of 40, the flow can be assumed to be quite turbulent at the inlet. However, even if the jets in the experiment appear as "ruffled" [7], side-view photographs indicate that the jets are still coherent when they impinge on each other. As mentioned in the Introduction, quite a different situation can be found in other studies [5] in which the jets pass each other without forming an impinging sheet. For the AR experiments [6,7], the effects of injection turbulence shall be examined in future work. In the present study, the exclusion of its effects isolates the role of fluid dynamic instability in sheet breakup.

Figure 6 displays the setup of the calculation and the diagnostics used to validate the high-velocity injection simulations. It shows that the Lagrangian particles are sampled across the plane at $z_0 = 20$ mm, that is, 16 mm downstream of the impingement point instead of the $z_1 = 45$ mm station reported by AR [6,7]. This choice is convenient in that it reduces the cell count on the base grid by two, and it can be justified when drops are completely converted into Lagrangian particles by the time they arrive at z_0. Because the Eulerian–Lagrangian transformation effectively "freezes" a liquid blob into a spherical drop, the particles passing z_0 are merely advected to z_1, and the time-averaged diameter distribution at the two planes is the same. This reasoning holds when, as it was verified in the simulation, only a small number of particles escapes through the y and x boundaries between z_0 and z_1. The main difference with respect to the measurement procedure by Anderson et al. [6] and Ryan et al. [7] is that the particles are collected across a 12×16 mm area to increase the number of samples available in the relatively short time spanned by the simulation, whereas PDPA interrogates a narrower area at the centerline. The assumption that the diameter distribution does not vary substantially in the sampling area is supported by the values of the arithmetic mean drop size (d_{10}) reported in [18] (Figs. 3–6) for $v_{\mathrm{jet}} = 18.5$. At $z = 20$ mm, the measured d_{10} is slightly less than 170 μm at the centerline and decreases to 145 μm at $y = \pm 6.4$ mm. At $z = 45$ mm, d_{10} is approximately constant and close to 175 μm. The small d_{10} variation (from 170 to 175 μm) between the two sampling stations confirms our assumption of similar average spray characteristics between z_0 and z_1.

Figure 6 also illustrates the position along the centerline of two pressure probes at $z = 12$ mm (probe 1) and $z = 14$ mm (probe 2). These probes will be used to assess the spacing of consecutive ligaments just after the sheet breakup. Because of surface tension, the pressure time history of the probes will increase in a gas-to-liquid pass and then decrease in a liquid-to-gas pass. Even if the curvature of the interface is unknown, these finite pressure pulses can be used to reveal the existence of a base frequency when the ligament formation is somewhat periodic.

In the following, we highlight results for three cases. The first, at the lowest injection velocity reported in the AR experiments [6,7], $v_{\mathrm{jet}} = 6.4$ m/s, is useful to verify the AMR block-structured algorithm as well as to validate the jets penetration length and the ligament spacing. The second and third, at $v_{\mathrm{jet}} = 12.2$ and 18.5 m/s, offer a more irregular liquid structure after impingement and, therefore, a greater uncertainty in evaluating these quantities. They also have a narrower spray distribution range compared with the low-velocity injection, which is better suited to verify the overall algorithm against the spray size distribution from PDPA measurements.

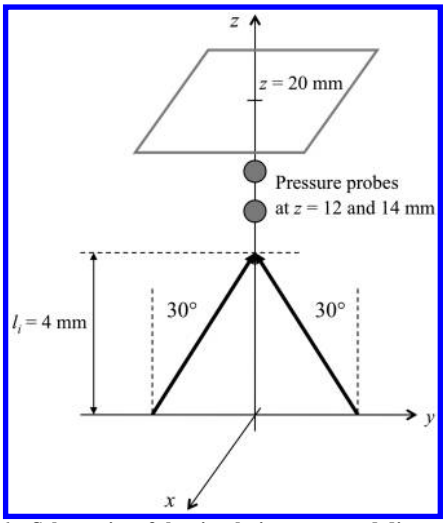

Fig. 6 Schematics of the simulation setup and diagnostics.

A. Low-Velocity Injection Case

The low injection velocity, $v_{\mathrm{jet}} = 6.4$ m/s, makes this case prone to the stretching of liquid membranes that eventually roll up and tear to form ligaments. To resolve these features, calculations are carried out with two and three levels of refinement starting from a coarse grid spacing of 125 μm. Figure 7 shows a snapshot of block distribution with the first and second levels of refinement added to the base grid: Fig. 7a depicts the x-z midplane section; Fig. 7b, the y-z midplane section; and Fig. 7c, the x-y section at $z = 1$ mm. Velocity vector fields from the base grid are also shown. The base grid squares and rectangles in each section are displayed in black, whereas the smaller AMR grid blocks are in gray. Note that the base grid is subdivided in blocks that may reside on different processors. This level of granularity of the AMR grid is controlled by the input maximum block size. As previously described, each level introduces a refinement by a factor of 2 with respect with to the previous (lower) level. By construction, the interface (the black thick line in the plot) is completely embedded by the highest level of refinement available. At later stages of the simulation, large drops (several multiples of the base grid spacing) form from fragments of the ligaments and advect downstream until they exit the computational domain. The AMR algorithm correspondingly builds fine grid blocks over most of the base grid.

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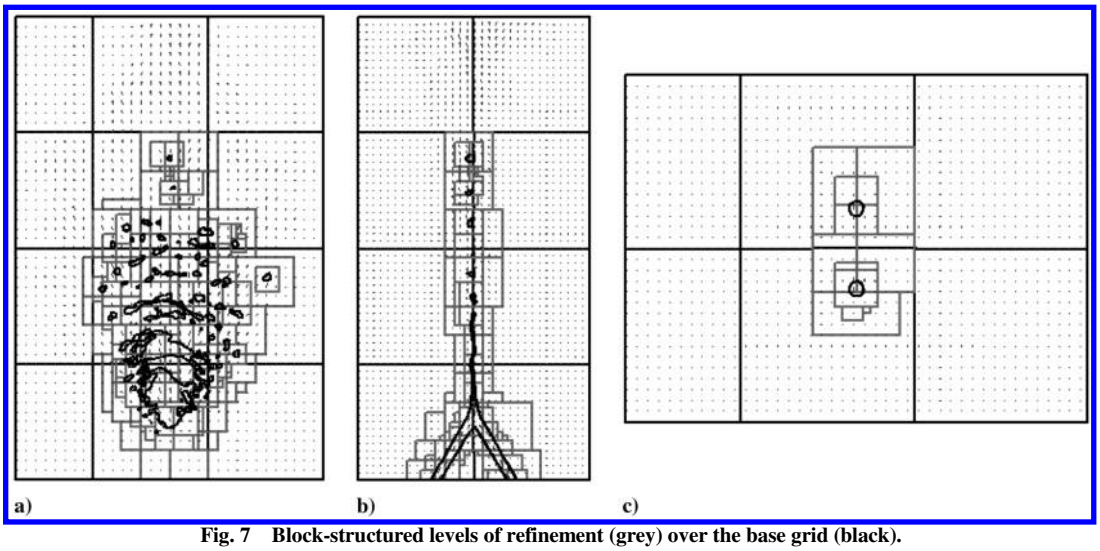

Fig. 7 Block-structured levels of refinement (grey) over the base grid (black).

A snapshot of the liquid sheet past the impingement point and from a direction perpendicular to the impingement plane is shown in Fig. 8 on the same length scale of the corresponding photograph in Ref. [7]. The plot illustrates how waves propagate radially from the impingement point, with tears along the intact sheet that deepen until they detach thin ligaments from the periphery of the sheet. The measured sheet breakup length in the AR papers is $x_b = 10 \pm 2.5$ mm. From the simulation, the average length estimated from 12 snapshots of the two-level simulation ($\Delta x_f = 31.25$ μm) is $x_b = 7.9$ mm. This value is at the low end of the measured range of core lengths. The addition of a third level of refinement ($\Delta x_f = 15.625$ μm) starting from the two-level simulation is accompanied by the appearance of more fine-scale structures in the impingement plane, but leads to a similar core length of approximately 8 mm. This result reflects the slow convergence of the interface-capturing algorithm in resolving a very thin membrane.

Further inspection of the time sequence of the rendered zero level-set isosurfaces suggests regular spacing of the thicker crests of the liquid membrane and the ligaments themselves. An "atomization frequency" is defined in the AR papers [6,7] as the ratio of sheet velocity to the spacing of detached ligaments, $f_a = v_{\text{jet}}/\Delta l$. From the calculation snapshots, we visually identify the maximum ligament spacing to be approximately 1.9 mm. The corresponding frequency $f_a = 3400$ HZ falls well within the error band reported for this injection velocity, because f_a is estimated from the AR plot to be between 2100 and 4000 Hz; see Table 1.

As an alternative to the previous visual definition, a rate of ligament formation can be found by analyzing the pressure signal from probes located immediately downstream of the sheet breakup point. Figure 9 illustrates the result of a grid convergence study of the pressure history at $z = 12$ mm for the duration of approximately a millisecond. It shows regularity in the spacing and amplitude of the signal peaks in the two- (dashed line) and three-level (continuous line) simulations. Of the two curves, the high-resolution calculation displays a tendency toward shorter peak spacing, that is, toward higher atomization frequency.

Table 1 Simulated impinging spray conditions[a]

v_{jet}, m/s	$d_0/v_{\text{jet}} \cdot 10^{-4}$, s	x_b/d_0	f_a, HZ	$x_{b,\text{calc}}/d_0$	$f_{a,\text{calc}}$, HZ
6.4	1	16 ± 2	3900(2000–4800)	13	3400
12.2	0.410	26 ± 4	5800(4600–7000)	20	4900
18.5	0.343	27 ± 4	6800(5700–8000)	16	4400–5900

[a]Data on the third and fourth columns, including the uncertainty intervals, are taken from the plots in [6].

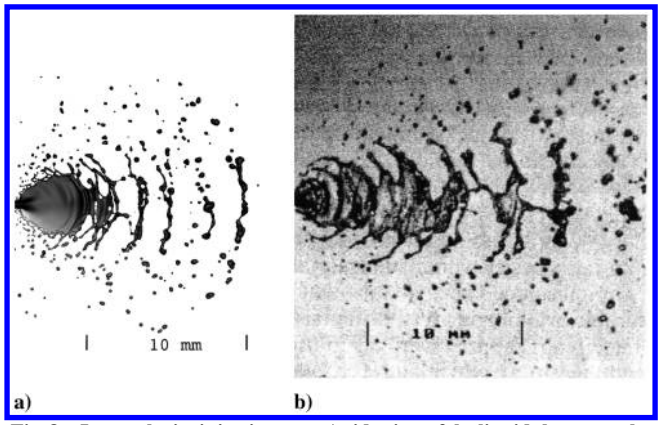

Fig. 8 Low-velocity injection case. A side view of the liquid sheet past the impingement point is shown from a) the simulation and b) the photograph reported in [7].

We can quantify these observations in the frequency domain by calculating the power spectral density (PSD) of the signal $\{p_i\}$, $i = 1, 2, \ldots, N$,

$$\text{PSD} = \frac{1}{N}\left|\sum_{i=0}^{N-1} p_i \mathrm{e}^{-j2\pi f_s i/N}\right|^2 \tag{15}$$

The PSD can be estimated with Welch's averaged modified periodogram method [19] by computing averaged periodograms of overlapped, windowed signal sections. From the two calculations, pressure values are taken consecutively (no subsampling) with sampling rates of $f_s = 7.7 \times 10^5$ and $f_s = 1.5 \times 10^6$, respectively, providing 1024 and 2048 sample points. The PSD is calculated with a Hanning window of size 512, and two points overlap. In the two-level simulation, the highest PSD intensity is found at $f_a = 2300$ HZ, whereas in the three-level simulation, the highest intensity is at $f_a = 2900$ HZ. Thus, it seems that the PSD analysis method is consistent with the visual exam of simulation images as both results point to an atomization frequency approaching 3000 Hz. However, the increase in estimated frequency with increased grid resolution suggests that the effect of grid resolution cannot be completely dismissed in the results.

We conclude this section by noting in Fig. 8 a qualitative agreement in largest droplet size between simulation and experiment. Because the size cutoff of the Lagrangian spray distribution must be comparable to the coarse grid spacing, only Lagrangian particles below 300 μm can form in this case, with most of the drops leaving the domain as Eulerian blobs. The quantitative comparison of droplet

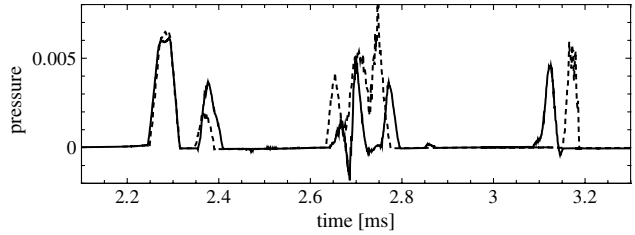

Fig. 9 Gauge pressure time history from probe at $z = 12$ mm: dashed line, $\Delta x_f = 31.25$ μm (two-level simulation); continuous line, $\Delta x_f = 15.625$ μm (three-level simulation).

size distribution is deferred to the next section, where higher injection velocities cause a narrower distribution range.

B. Higher Injection Velocities

We consider first the case $v_{\text{jet}} = 18.5$ m/s. Figure 10 shows snapshots of the liquid sheet past the impingement point from calculations with increasing grid resolution. These can be compared in Fig. 10d with a photograph of the sheet taken at the same conditions and on the same length scale. With $\Delta x_f = 62.5$ μm (one level of refinement), a rather modest liquid sheet forms, with a few ligaments. Fig. 10a shows that the sheet is substituted by large particles due to the conversion in Lagrangian form. Such a low grid resolution appears to be insufficient for capturing the basic atomization mechanism of sheet formation and breakup. The addition of one refinement level ($\Delta x_f = 31.25$ μm) leads to a substantial improvement in the appearance (in Fig. 10b) of the liquid sheet, which is now longer and broader. By adding one more refinement level ($\Delta x_f = 15.625$ μm), the impingement sheet still evolves and reveals longer and more arched ligaments (Fig. 10c), similar to the photograph of the AR experiment at these same conditions from Ref. [7]. This photograph is displayed in Fig. 10d.

Compared with the $v_{\text{jet}} = 6.4$ m/s case, the ligaments still display a somewhat regular spacing near the core, whereas droplets tend to disperse in an apparently more disorganized pattern in the far field. Similarly, the sequence of pressure peaks from the time histories of the two probes, shown in Fig. 11, is very noisy, but suggests a measurable periodicity. To decrease the subjectivity in the estimate of the atomization frequency, we abandon the visual inspection of ligament spacing adopted by AR [6,7] and rely instead on the PSD analysis described in the preceding section. In the frequency domain (Fig. 12), the largest signal intensity is found at 5900 Hz from probe 1 and at 4400 Hz from probe 2. The dispersion in the results confirms a less-organized pattern of atomization.

The peak intensity frequencies from the two probes are at the lower end of the 6000–8000 Hz range from the AR experiments [6,7]. It is possible that the actual frequencies are even higher: The value

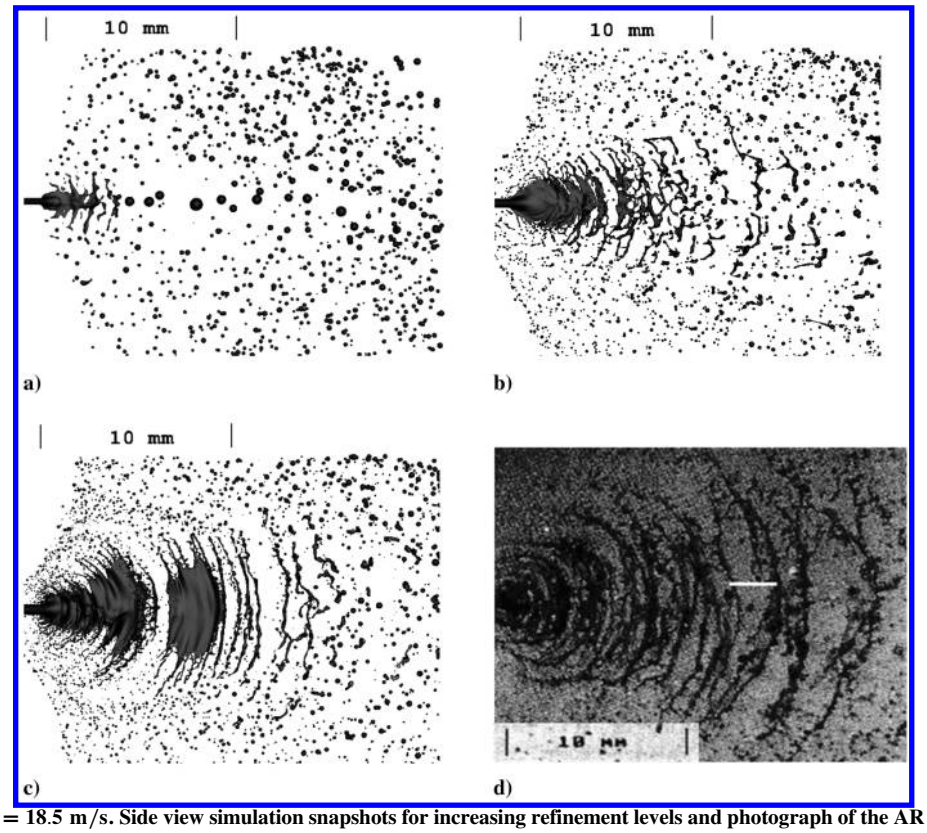

Fig. 10 Case of $v_{\text{jet}} = 18.5$ m/s. Side view simulation snapshots for increasing refinement levels and photograph of the AR experiment from [7].

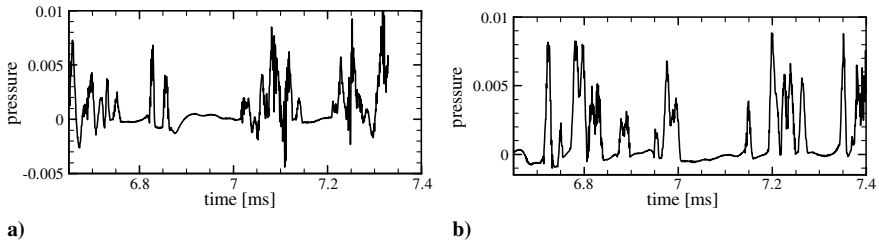

Fig. 11 Pressure time history at two centerline locations past film breakup: a) $z = 12$ and b) $z = 14$ mm.

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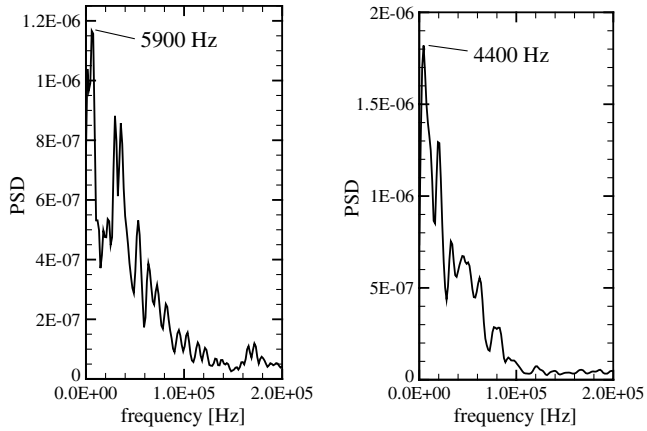

Fig. 12 Power spectral density of the two pressure signals from the $v_{\text{jet}} = 18.5$ m/s simulation at a) $z = 12$ and b) $z = 14$ mm. The sampling frequency is 1.5×10^6 HZ.

$f_a = 7000$ HZ reported in the experiment corresponds to $\Delta l = 2.6$ mm. This length, marked as a white segment on the snapshot of Fig. 10d, seems larger than the ligament spacing there, and stresses how the visual recognition of features based on the perception of continuity can be somewhat subjective. Figure 12 indeed indicates substantial intensity peaks above 10 kHz, especially at $z = 12$ mm. Similarly, the core length reported by AR is $x_b = 17.5 \pm 2.5$ mm, almost twice the length of the low-velocity injection case. On the snapshot of Fig. 10d, this length extends from the impingement point to about two-thirds of the field of view, spanning what could possibly be disconnected ligaments. If these closely generated ligaments are included, the intact sheet length that is evaluated from the simulation is approximately 10 mm, possibly 11 mm, therefore at the low end of the AR uncertainty interval. No substantial increase of the average core length is observed between the intermediate and the highest resolution calculation attempted in this case.

The similarity of spray dispersion in Figs. 10b and 10c is supported by the analysis of the normalized histograms of droplet size in the far field. All the droplets are converted into Lagrangian particles before exiting the domain, making the comparison with PDPA data very simple. The distribution at $z = 20$ mm from three simulations with increasing grid refinement is compared in Fig. 13 with the spray PDPA measurements taken at $z = 45$ mm. The justification for the different location of the sampling planes was previously explained in terms of almost identical size composition in this more dilute spray region. For $\Delta x_f = 31.25$ μm, droplets were sampled between 5.24 and 6.58 ms for a total of 1086 samples. A lower number of samples (480) is currently available for the $\Delta x_f = 15.625$ μm calculation due to the much higher cost involved in carrying out the simulation.

We see from the histogram in Fig. 13a that the droplets from the lowest resolution case are too large, with a peak in frequency around 200 μm, whereas the intermediate (Fig. 13b) and fine (Fig. 13c) grid resolutions are in good agreement with the data.

The last case considered in this work is $v_{\text{jet}} = 12.2$ m/s. As before, two and three levels of refinement were used to examine the convergence of the solution. The lower injection velocity results in longer lasting ligaments with respect to $v_{\text{jet}} = 18.5$ m/s, as well as in a broader spray distribution. The snapshot of the three-level calculation in Fig. 14 suggests that a few ligament fragments leave the domain as liquid blobs with diameters larger than 300 μm. No photograph is available for this case. The atomization frequency is extracted from 8192 pressure samples in a two-level calculation pressure with Hanning window of size 4096 and two points overlap. From probe 2, $f_a = 4900$ HZ, which is comparable with the measured interval 4600–7000 Hz.

The core length reaches values of 14–15 mm (for instance, in the snapshot of Fig. 14a) in the three-level calculation. This value is shorter than $x_b \sim 16.5$ mm reported in the AR experiments [6,7]. Finally, the spray size distribution at $z = 20$ mm calculated from the two-level calculation with 1850 samples is shown in Fig. 14b. It correctly captures the most frequent particle diameter at 120 μm, but misses many of the droplets smaller than 80 μm. Drops larger than 300 μm were identified separately on the grid as isolated blobs (200 samples); their effective diameter (the diameter of the sphere with the same volume of the Eulerian blob) is also included in the distribution.

IV. Trends from Linear Instability Analysis

The observed periodic nature of liquid sheet breakup suggests that fluid dynamic instability may be the dominant process in this atomization device. Linear temporal stability analysis can be used to predict the fastest growing mode via a dispersion relation that includes the effect of aerodynamic forces. High-fidelity calculations can assist by providing values of reference thickness and sheet velocity. In examining slices cut along the midplane of the computational domain (Fig. 15), two observations can be made: The sheet velocity is close to the injection value, and the sheet thickness does not decrease substantially before breakup.

Clearly visible in Fig. 15 are also long-wave sinusoidal oscillations. The fastest growing oscillation can be deduced for an inviscid, constant thickness, infinite sheet according to the well-known expression for the complex angular frequency in the dimensionless constant thickness $h = 2t/\lambda$ [20],

$$\omega = \frac{\tan\, h(2\pi h) - \rho}{\tan\, h(2\pi h) + \rho}\pi \pm 2\pi\left(\frac{\rho}{1+\rho}\right)^{1/2} \times \frac{\{W[\tan\, h(2\pi h) + \rho] - (1+\rho)\tan\, h(2\pi h)\}^{1/2}}{\tan\, h(2\pi h) + \rho} \tag{16}$$

where $\rho = \rho_G/\rho_L$ is the gas–liquid density ratio, and

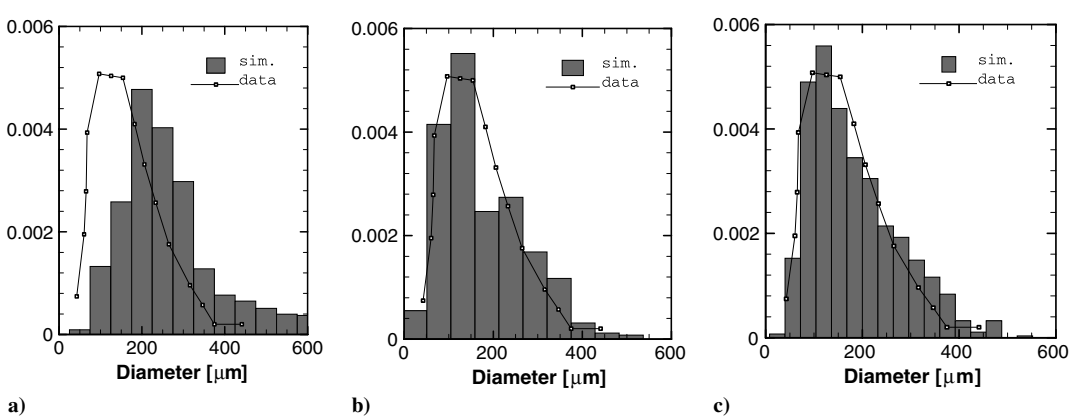

Fig. 13 Case of $v_{\text{jet}} = 18.5$ m/s. Comparison of the droplet size distribution from the AR experiments [6] (symbols) and from simulations with increasing levels of refinement.

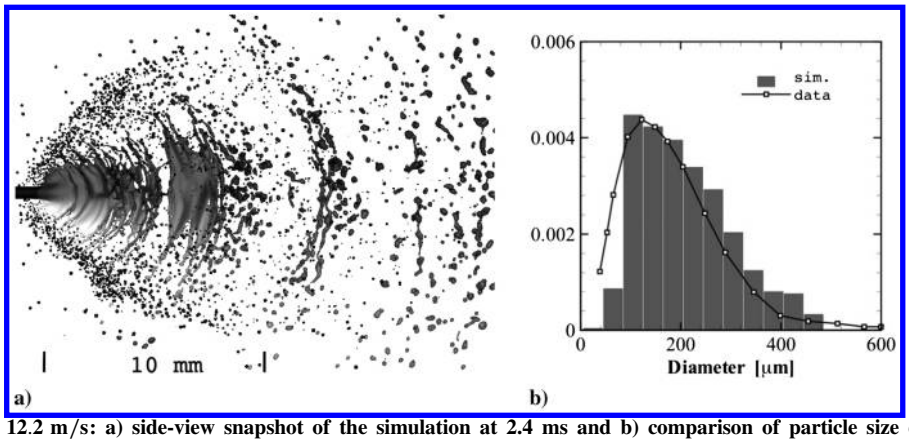

Fig. 14 Case of $v_{\text{jet}} = 12.2$ m/s: a) side-view snapshot of the simulation at 2.4 ms and b) comparison of particle size distribution with PDPA measurements.

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$$W = \frac{2\pi}{\lambda(\Delta U)^2}\left(\frac{1}{\rho_L} + \frac{1}{\rho_G}\right) \tag{17}$$

is the normalized wave number corresponding to the perturbation wavelength λ. Taking the velocity difference across the interface to be $\Delta U = v_{\text{jet}}$, the frequency of the most unstable mode is 162, 1180, and 4170 Hz for $v_{\text{jet}} = 6.4$, 12.2, and 18.5 m/s, respectively. The half-thickness is $t = 0.1$ mm in all the cases; even variations of ± 0.05 mm in t only produce small percentage variations in frequency.

The values derived from linear instability analysis are substantially smaller than the atomization frequencies of the AR experiments [6,7], particularly the frequency at the low injection velocity. However, they capture the correct trend of increasing f_a with increasing v_{jet}, suggesting that the Kelvin–Helmholtz (KH) instability [21] may indeed be at the base of ligament formation. Making KH frequency predictions more quantitative requires the revision of some of the assumptions that lead to Eqs. (16) and (17). According to these equations, the sensitivity to t is very small at the density ratio of the AR experiments, therefore, the correction by Dombrowski and Johns [22] for a thinning sheet can be considered a second-order effect. More relevant to correct the KH analysis estimates is perhaps the actual thickness of the boundary layer that forms on the gas side of the liquid sheet. Although waves from a thin boundary limit could form close to the impingement point, where a boundary layer has no time to form, a thick boundary may provide the dominant length scale further downstream. This gas boundary-layer effect has emerged in recent analysis of liquid jet in crossflow atomization at low to moderate Weber numbers [23]. In this regime, the need to characterize the boundary-layer thickness along the impingement sheet justifies further use of high-fidelity simulations. For instance, the increase in atomization frequency with increased grid resolution in the convergence study of Fig. 9 could be attributed to the sharper velocity profiles across the liquid surface.

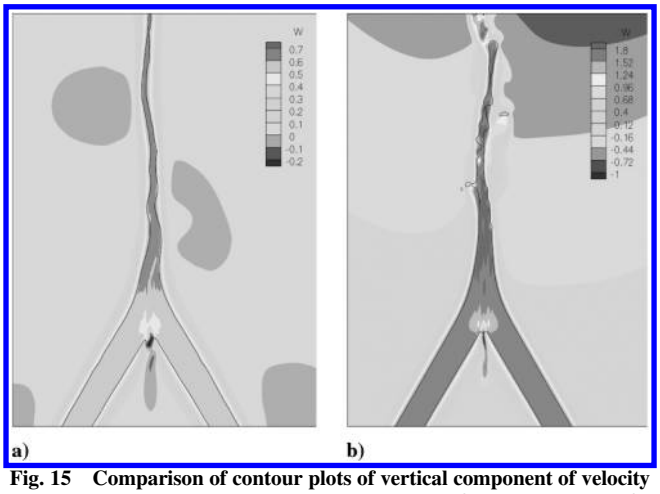

Fig. 15 Comparison of contour plots of vertical component of velocity (w) along the center plane slice in units of 10 m/s; a) $v_{\text{jet}} = 6.4$ m/s, b) $v_{\text{jet}} = 18.5$ m/s.

V. Conclusions

This study establishes the combined level-set/volume-of-fluid methodology as a tool for predicting spray atomization without empiricism (other than the standard transport model for isolated Lagrangian droplets). The salient features of the spray fan breakup and ligament frequencies from this method show overall agreement with both trends and magnitudes of the experimental results. Effects due to limited grid resolution cannot be completely ruled out in the results, which consistently underestimate the core length of the impingement sheet. However, the spray droplets distribution is observed to converge to the phase Doppler particle analyzer data relatively quickly. To the authors' knowledge, this is the first instance of a first-principles primary atomization calculation executed on the scale and at the conditions of an actual experiment and directly validated against spray measurements. All the calculations are carried out using a small subset of a modern computer cluster.

The simulations and the experiments for this injection configuration point to the dominant role of the liquid sheet formed from jet impingement. Especially at low injection velocity, measurements of ligament spacing (both visual from simulation snapshots and in the frequency domain from pressure histories of pointlike probes) show significant periodicity, consistent with the hypothesis of onset of fluid dynamics instability of the Kelvin–Helmholtz type. A simplified linearized analysis confirms the trend of higher rate of ligaments generation with increasing injection velocity. Effects due to the turbulent state of the liquid phase at the injection orifice are completely neglected in this study, but it can be expected that, when the jets are partially or completely disintegrated before impinging, fluid dynamic effects due to details of the specific injector device geometry will compete with long-wavelength fluid dynamics instability in determining the dominant mechanism of spray formation.

In closing, the authors stress that the Eulerian-to-Lagrangian transformation can enable simulations over relatively large domains if it is carried out in the context of block-structured adaptive mesh refinement. The limits of this hybrid approach are dictated by the grid spacing and the number of refinement levels that can be afforded in the simulation. The first parameter, Δx_c, controls the size of the largest Lagrangian particles; it cannot be too large because it also fixes the cutoff wavelength of turbulence in the bulk gas phase. The total number of refinement levels determines Δx_f, which in turn establishes the minimum droplet size that can be resolved. The number of affordable levels is currently small because of the overhead introduced by managing many thousands of computational blocks in the grid hierarchy. However, work is in progress to improve the performance of the multigrid preconditioned conjugate gradient method to accelerate the convergence of the pressure equation on the hierarchy of grids. This new algorithm is expected to preserve the benefits and convergence properties of the original multigrid

preconditioned conjugate gradient method solver while reducing the number of computational grid points required by uniform grids, therefore allowing smaller blocks and larger numbers of refinement levels.

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